

projects\nas-sos-capstone\cameo_models\nas_context_diagram.sysml

```
1 package ContextDiagram {
2   doc /* Context diagram for the NAS SOS Capstone project, showing the system of
3     * interest (SoI), the external entities, and the exchanges between them.
4     *
5     * Per D-007 the SoI is the NAS made of the five modeled domains (Airspace
6     * Management, Airport Operations, Flight Operations, Aircraft Systems, Flight
7     * Crew). Outside it, in Environment: boundary actors (Governance, Passengers,
8     * Military) and context constraints (Infrastructure / Airspace Resources,
9     * Maintenance Suppliers). Information Services and Decision Support are absorbed:
10    * they appear only as exchange content, media, or capabilities.
11    * Per D-009 only modeled systems are rendered (#ContextVisible) for now.
12    * Authoring rules: knowledge/models/sysml-modelling-methods.md §3, §10.
13    */
14
15   metadata def ContextVisible {
16     doc /* Tag for elements the context view (MyViews::systemContext) should render.
17       * Apply as a prefix: #ContextVisible part ... / #ContextVisible connect ... */
18   }
19
20   part def NASContextDiagram {
21     private import ScalarValues::*;
22     private import NationalAirspaceSystem::AirportOperations::*;
23     private import NationalAirspaceSystem::AirspaceManagement::*;
24     private import NationalAirspaceSystem::AircraftSystems::*;
25     private import NationalAirspaceSystem::FlightOperations::*;
26     private import NationalAirspaceSystem::FlightCrew::*;
27     private import NationalAirspaceSystem::Common::*;
28     private import NationalAirspaceSystem::Environment::*;
29
30     // #ContextVisible marks what the context view renders. Nested parts are
31     // redefined (:>>) only so they can carry the tag; their types are unchanged.
32     //
33     // D-007 assessment (D-009, decisions-log.md). Every system here is tested against
34     // D-007's rule: MODELED if it has distinct decision authority or execution behavior,
35     // dynamically coupled to in-scope decisions, that materially affects cost, safety,
36     // workload, schedule reliability, or passenger value. Otherwise it's a BOUNDARY ACTOR
37     // (supplies objectives/demand/feedback/rules), a CONTEXT CONSTRAINT (exogenous), or
38     // ABSORBED (kept only as exchange content or capability). Rendered = on the context view.
39
40     // D-007: BOUNDARY ACTORS (governance, passengers, military), CONTEXT CONSTRAINT
41     // (infrastructure), ABSORBED (informationServices, decisionSupport). Not rendered yet;
42     // links drafted in "MISSING". Maintenance Suppliers (context constraint) has no part yet.
43     part environment: EnvironmentDomain;
44     #ContextVisible part airportOps: AirportOperationsDomain {
45       // D-007: MODELED. Authority over gates, non-movement area, closures, and slots at
46       // coordinated airports; departure readiness couples to EDCT compliance and
47       // schedule reliability. Weakest evidence of the modeled set (C/B). Rendered.
48       // Its sub-parts (fuel, ground handling, gates, terminals, baggage, cargo,
49       // maintenance) are ABSORBED: they carry out decisions made elsewhere (owner
50       // 2026-09-27), so they appear only as ends of the unrendered aircraft <-> airport set.
51       #ContextVisible part :>> airport;
52     }
53     #ContextVisible part airspaceMgmt: AirspaceManagementDomain {
54       // D-007: MODELED. Tactical control authority (clearances, separation; JO 7110.65BB)
55       // directly shapes the flown trajectory; safety + schedule. Rendered.
56       #ContextVisible part :>> atc;
57       // D-007: MODELED. "Final approving authority" for interfacility TMIs (7210.3EE
58       // 18-2-3i); EDCT/CTOP/reroutes couple straight to airline decisions; delay cost and
59       // schedule reliability. Rendered.
60       #ContextVisible part :>> atccsc;
61       // D-007: MODELED. Local TM attitude (MIT, fix balancing; 18-2-3 NOTE); turns
62       // national TMIs into constraints on the sectors; schedule + controller workload. Rendered.
63       // One part stands for terminal and ARTCC TMUs.
64       #ContextVisible part :>> tmu;
65       // tracon / artcc: MODELED in the definitions, but at context level they are covered
66       // by the atc umbrella (see the atc TODO below). Not rendered.
67     }
68     #ContextVisible part aircraftSys: AircraftSystemsDomain {
69       // D-007: MODELED. No decision authority, but its execution behavior (FMS intent ->
70       // guidance -> motion) IS the end of the intent chain; fuel burn, safety. Rendered.
71       #ContextVisible part :>> aircraft;
72     }
73     #ContextVisible part flightOps: FlightOperationsDomain {
74       // D-007: MODELED. Operational control; the dispatcher and PIC are jointly
75       // responsible for the release (14 CFR 121.533); sets the airline's plan and fuel;
76       // cost and schedule. Rendered.
77       #ContextVisible part :>> airlineOperationsCenter;
78       // ticketingSystem: ABSORBED. Revenue/booking side; enters only as passenger demand
```

Add note about how AirspaceMgmt encompasses a number of sub-systems.



Add note about sub-sys? See evidence source for RA, ATC crew, etc. Is this in part def? If not, add it there & comment here.

```
79 // (a boundary-actor effect per D-007). Not rendered.
80 }
81 #ContextVisible part flightCrew: FlightCrewDomain {
82 // D-007: MODELED, as ONE element (D-009 merge). The PIC holds final authority
83 // (14 CFR 91.3) and turns the plan + clearances into FMS intent; safety,
84 // workload, fuel. The PIC / FO split is not a separate authority at context
85 // level, so the two roles live inside flightDeckCrew and aren't rendered.
86 #ContextVisible part :>> flightDeckCrew;
87 // cabinCrew: inside the MODELED Flight Crew domain but ABSORBED at context level.
88 // No authority over in-scope decisions; "cabin secure" couples only weakly to
89 // departure time, through the PIC. Not rendered (owner, 2026-09-27).
90 part :>> cabinCrew;
91 }
92
93 // -----
94 // REVIEW KEY (added 2026-09-27)
95 // [A] primary/normative source states it (JO 7110.65BB, JO 7210.3EE)
96 // [B] empirical literature shows it (Seamster 2011, Berry & Pace 2011, Munro 2018)
97 // [C] your notes / general domain knowledge; defensible, not yet citable
98 // TODO(review) no evidence, a direction/term problem, or weak tie to the
99 // intent chain (index.md "Center of gravity") - your call
100 // TODO(missing) something the notes leave out; draft code is commented out
101 // in the "MISSING" section at the bottom
102 // Evidence detail per connection: knowledge/models/context-diagram-exchange-evidence.md
103 // -----
104
105 // Theorize what we want to connect up here.
106 // The ATM consists of:
107 // - The aircraft:
108 //   - flying from airport to airport, creating 4D points that form a trajectory,
109 //   - controlled by the flight crew, physical interactions
110 //   - observed by the airline operations center,
111 //   -> [B] implemented below as aocAircraftDatalink (ACARS/OOOI). Was not drawn before.
112 //   - observed by the airspace management domain
113 //   -> [C] implemented as atcSurveillance.
114 //   - interacting with the environment domain (weather, terrain, etc.)
115 //   -> TODO(missing): environment is declared but has no connections. Draft below.
116 //   - operated by the Flight Operations domain,
117 //   - certified according to the rules established by the environment domain.
118 // - The flight crew:
119 //   - supported by the flight operations (e.g. airline operations center),
120 //   - interacting with the aircraft systems domain,
121 //   - interacting with the airspace management domain,
122 //   - interacting with the airport operations domain,
123 //   - working according to the rules established by the environment domain.
124 // - The flight operations domain:
125 //   - supporting one to many flight crews,
126 //   - receiving data from the aircraft systems domain,
127 //   -> [B] aocAircraftDatalink below.
128 //   - communicating with the airspace management domain,
129 //   - communicating with the airport operations domain.
130 //   -> TODO(missing): the connection exists (aocAirport) but there is no
131 //     exchange list for it in your notes. See the aocAirport doc.
132 //   - working according to the rules established by the environment domain.
133 // - The airport operations domain:
134 //   - communicating with the flight operations domain,
135 //   - communicating with the airspace management domain,
136 //   - sending data to the aircraft systems domain,
137 //   -> RESOLVED 2026-09-27: mostly material, some data. Full set is kept at the
138 //     aircraftAirport connection, rendered as one summary link (no effect on
139 //     optimality; driven by other actors).
140 //     Airport lights and signs don't count toward this link either way; they
141 //     are visual cues to the crew, not data to aircraft systems:
142 //     - Airport lighting isn't an ATC -> crew exchange. The tower controller
143 //       operates the lights (ATC -> airport, BB 3-4-2..3-4-15), and the lights
144 //       signal the pilot (airport -> crew). Those halves live on atcAirport
145 //       and airportToCrew.
146 //     - Runway and taxiway signs are fixed (installed and maintained by the
147 //       airport operator), so ATC doesn't drive them.
148 //     What airport -> aircraft data remains is navaid signals (ILS), and D-007
149 //     may put those under Infrastructure instead.
150 //   - communicating with the flight crew,
151 //   - working according to the rules established by the environment domain.
152 // - The airspace management domain:
153 //   - communicating with the flight operations domain,
154 //   - communicating with the airport operations domain,
155 //   - sending data to the aircraft systems domain,
156 //   -> TODO(review): direction is mostly backwards. Surveillance data runs
157 //     aircraft -> ATC (transponder / ADS-B). ATC's instructions reach the aircraft
158 //     only through the crew (clearances), or via datalink (PDC over ACARS,
```

also are primary interface to passengers, so they deal w/ satisfaction & relay to/from flight deck.

Note FMS-Pilot shared control won't be modeled, but may prove important? On modern ALC, FMS can't act w/o pilot intervention via MCP@min.

Reviewed? ☒

☒

is "Datalink" misused vs. std. industry terminology? Here I mean to include ADS-B w/ ACARS/CPDLC. Broken out in parts def?

☒ Flight Ops to Airport: - into flowing through gate personnel? - gate block on/off times? Need

Source - Runway closures, fuel costs & availability?

☒ ILS, navaids maintained by FAA? Similar to Lights then.

This is through the alt systems!

```
159 // CPDLC) that the crew accepts and enters.No ATC -> aircraft control path.
160 // - communicating with the flight crew,
161 // - working according to the rules established by the environment domain.
162
163 // I need to connect only the relationships that actually matter operationally,
164 // not every domain to every other domain.
165 //
166 // The anonymous `connect` statements that were here are now the named, documented
167 // connections below, one after each note block that justifies it. Same ends, same
168 // #ContextVisible tag, so the view renders the same links, now labelled.
169
170
171 // Core operational coordination - AOC to Crew
172 // - Recommended fuel loading & anticipated passenger load
173 // -> [B] Dispatch release (Seamster E-1/E-2; Munro 2018). TODO(review) wording:
174 // fuel isn't recommended. The release sets fuel that the dispatcher AND
175 // captain both sign; the captain may ask for more (Munro). The passenger load and
176 // final W&B come from load control (Seamster E-2#20).
177 // - Flight plan & route info (SID/STAR, altitudes, waypoints, etc.) -> [B] E-1/E-2
178 // - Weather & NOTAM info -> [B] release package
179 // -
180 // -> TODO(missing): the empty bullet. Candidates with [B] evidence: the dispatch
181 // release itself as one object (distinct from the ATC flight plan, Munro);
182 // alternates and MEL items; amended release / reroute + new ETA after a TFM
183 // reroute (E-7#11-13); EDCT relayed from TFM.
184
185 // Core operational coordination - Crew to AOC
186 // - Final Passenger & Fuel Load
187 // -> Intent (owner, 2026-09-27): the crew's higher-resolution view of the fuel
188 // actually loaded goes back to the AOC. Fuel is metered in gallons, then converted
189 // to pounds using the fuel density. The crew checks that figure against the
190 // fuel gauges and against the planned fuel on the release.
191 // [B] crew -> AOC fuel on board (E-7#08-09, E-9#06-09); release acceptance /
192 // requests for more fuel (Munro). [C] the fuel slip and gauge cross-check are
193 // standard practice but not yet in a registered source. Add?
194 // Passenger count and final W&B still flow load control -> crew (E-2#20, ACARS);
195 // the crew only reports back a discrepancy, not the count.
196 // RESOLVED (owner, 2026-09-27): the airport's fuel carrier fuels
197 // (Airport.fuelDistributionSystem). The fuel slip reaches the crew on
198 // airportToCrew; the uplift itself is apFuelUplift / apRefuelPanel. No
199 // maintenance link is needed for fuel.
200 // TODO(review), minor: does the uplift record ALSO go straight to the AOC
201 // (electronic fueling records, [C])? Only matters if the AOC reconciles fuel
202 // without waiting for the crew's report.
203 // - Aircraft Configuration & Performance Data
204 // -> Intent (owner, 2026-09-27): the FMS computes its own performance numbers,
205 // and the crew reviews them against the dispatch paperwork.
206 // FMS side (aircraft -> crew): predicted fuel at destination and alternate, ETAs,
207 // gross weight (from the ZFW + fuel the crew entered), optimum/max altitude, and
208 // on some types takeoff speeds. Paperwork side (AOC -> crew): planned burn, times,
209 // ZFW / takeoff weight, cost index. This is the point where "flight plan" becomes
210 // "FMS intent" in the intent chain. The cross-check is the only place anyone
211 // checks that the two agree, so it's worth modeling explicitly.
212 // [C] standard flight-deck practice; FMS initialization inputs in ConOps Phase 3.
213 // TODO(review) evidence: needs an FCOM / operator procedures source for the
214 // cross-check and its tolerances.
215 // RESOLVED (owner, 2026-09-27) abstraction: takeoff performance (V-speeds,
216 // thrust / assumed-temperature settings) is modeled as AOC-provided, part of
217 // the release package on aocToCrew. In reality it depends on the aircraft type:
218 // some types take it from the airline, others compute it in the FMS (e.g. the
219 // 737). The FMS side of the cross-check is therefore limited to fuel, weight,
220 // ETA, and altitude predictions.
221 // What crew -> AOC carries is the OUTCOME: acceptance, or a discrepancy (fuel,
222 // weight, performance limit) that needs an amended release. The comparison itself
223 // happens inside the crew, drawing on two links: aocToCrew (release) and
224 // crewAircraft (FMS predictions). That's why it's a behavior (an action
225 // def), not a third connection. Draft in "MISSING".
226 // The dispatch paperwork is the dispatch release package: the item that carries
227 // the plan, delivered on paper, EFB or ACARS. See the AOC to Crew TODO(missing)
228 // above; draft item def in "MISSING".
229 // If you meant MEL/configuration deviations instead, those belong under "status
230 // & maintenance" below.
231 // - Aircraft Status & Maintenance Needs -> [B] to maintenance control (Berry & Pace);
232 // often automated, i.e. aocAircraftDataLink
233 // - Equipment Malfunctions & Emergencies -> [B] Seamster T2.10 (diversion, emergency)
234 // - Pilot Reports (PIREPs) -> [C] common practice (ride reports to
235 // dispatch); the sources only cover PIREPs to ATC
236 // - Route Modification Needs -> [B] Seamster A.1#41-46 (reclearance,
237 // route change, ETA change > 5 min)
238 // -> TODO(missing): ETA changes and unplanned holding (A.1#41-46); diversion notification.
```

Should I add non-visible
other, embed Fuel & PAX in
Dispatch paperwork,
or omit others?
Think D-007.

Discrepancies. Change.

No idea. Let's say uplift goes
through crew to AOC b/c not all
airports hi-tech enough.

Tolerances don't
seem particularly important
yet. Ignore for now.

Condense this
Commentary.

These fall into Route Mods
to me.

```
239 // OOOI times are NOT crew -> AOC: they're automated avionics reports (aocAircraftDataLink).
240
241 #ContextVisible connection aocToCrew connect flightOps.airlineOperationsCenter to flightCrew.flightDeckCrew {
242   doc /* [B] Dispatch/operational control link (dispatcher <-> flight deck; the captain signs).
243     * AOC -> crew: dispatch release (route, fuel, alternates, weather, NOTAMS, MEL),
244     *   amended release / reroute + ETA, load & final W&B (from load control).
245     * Crew -> AOC: release acceptance / fuel requests, fuel on board (after the
246     *   fuel-slip / gauge check), release-vs-FMS discrepancies, maintenance
247     *   and malfunction reports, diversion/emergency, route-change needs, ETA changes.
248     * Media: phone radio pre-departure; ACARS/Satcom airborne. Both sign the release; the
249     *   captain has final authority (14 CFR 91.3).
250     * Intent chain: mission plan -> flight plan (the airline's intent reaching the crew).
251     * Sources: Seamster 2011 E-1, E-2, E-7, E-9, T2-10, AMA; Munro 2018; Berry & Pace 2011. */
252 }
253 // RESOLVED (D-009, 2026-09-27): PIC and FO merged into flightCrew.flightDeckCrew. Every
254 // crew link was drawn twice with identical content, and D-007 finds no separate
255 // decision authority between the two pilots at context level. The "FO works the radio"
256 // detail moves to the IBD, where both roles still exist as sub-parts.
257
258 // Core operational coordination - ATC to Crew
259 // - radio comms hand-offs & transfer info -> [A] BB 5-4-5..5-4-8, 2-1-15; Seamster E-6#03-04
260 // - traffic advisories -> [A] BB 2-1-21
261 // - request responses/clearances -> [A] BB 4-2-x, 4-3-x, 3-7-2, 3-9-x, 4-8-1
262 // - ATIS info -> [A] BB 2-9-2/3 (broadcast)
263 // - Coordinate airport lighting to communicate to pilots statuses & preferences
264 //   -> TODO(review): not an ATC -> crew exchange. BB 3-4-2..3-4-15: the tower
265 //   controller OPERATES the lighting (ATC -> airport equipment), then the lights
266 //   signal the pilot visually (airport -> crew). You already have both halves in
267 //   the "ATC to Airport" and "Airport to Crew" notes, so delete this line.
268 //   Lights don't carry "preferences"; pilots can ASK for intensity changes.
269 //   Also weakly tied to the intent chain (surface safety, not trajectory intent).
270 // - Estimated Departure Clearance Times (EDCTs)
271 //   -> [A] BB 4-3-4 (hold-for-release / EDCT in the clearance, E-2#10).
272 //   TODO(review) term: EDCT = "Expect Departure Clearance Time" (FAA P/CG,
273 //   glossary.md). The ATCSCC assigns it; ATC only relays and enforces it
274 //   (-5/+5 min). The crew usually gets it from dispatch first.
275 //   -> TODO(missing): holding instructions + Expect Further Clearance (BB 4-6-1,
276 //   Seamster E-9); safety alerts (terrain/traffic); altimeter settings; approval of
277 //   weather deviations; PIREP solicitation (BB 2-6-2). Holding and deviations are
278 //   trajectory-changing, so they matter for the intent chain.
279
280 // Core operational coordination - Crew to ATC
281 // - Intent, Requests, & Readbacks & Reports -> [A] BB; Seamster T2.7#21, T2.4#06
282 // - Equipment Malfunctions & Emergencies -> [A] BB 2-1-7, Ch. 10
283 // - Pilot Reports (PIREPs) -> [A] BB 2-6-2
284 // - Comms Establishing at departure and in response to transfer
285 //   -> [A] BB 2-9-3 (ATIS code on contact); Seamster E-5#08-10
286 // -> TODO(missing): read-backs (the crew's acceptance of a clearance, i.e. where
287 //   "cleared" becomes "flown"); position/altitude reports; ICAS RA and TAWS reports
288 //   (BB 2-1-28/2-1-30: ATC separation responsibility is suspended during an RA);
289 //   "unable" responses.
290
291 #ContextVisible connection atcToCrew connect airspaceMgmt.atc to flightCrew.flightDeckCrew {
292   doc /* [A/B] Tactical control: the best-evidenced link in the model.
293     * ATC -> crew: clearances (IFR/PDC, taxi, takeoff, altitude/heading/speed/route,
294     *   approach, landing), frequency changes/handoffs, traffic advisories, ATIS,
295     *   EDCT/hold-for-release as relayed from TFM.
296     * Crew -> ATC: requests and intent, read-backs, check-in with ATIS code,
297     *   malfunctions/emergencies, PIREPs.
298     * Media: radio (145 of ~277 Seamster matrix rows); PDC via ACARS.
299     * Authority: ATC holds tactical control; clearances leave "little room for
300     *   negotiations" (Seamster p.46); the PIC keeps final authority (14 CFR 91.3).
301     * Intent chain: ATC constraints -> trajectory negotiation.
302     * Either pilot may be the voice on the radio (the FO in E-2#07, E-4#02-04).
303     * Sources: JO 7110.65BB (2-1-21, 2-6-2, 2-9-2/3, 4-3-4, 5-4-5..8); Seamster E-1..E-13. */
304 }
305
306 // Core operational coordination - ATC to AOC (ATSCC -> Flight Ops)
307 // - Traffic Advisories
308 //   -> [A] content is right, term is wrong. TODO(review): "traffic advisory" is
309 //   the BB 2-1-21 controller -> pilot proximity call. The ATCSCC sends
310 //   operators ADVISORIES (GDP, GS, AFP, CTOP, reroute; 7210.3EE 18-10-6c,
311 //   18-12-4, 18-15-2a7). Rename to "TMI / reroute advisories".
312 // - Estimated Departure Clearance Times (EDCTs)
313 //   -> [A] 7210.3EE 18-10-6e: EDCTs go "to ARTCCs and linked system users".
314 //   (Same "Expect" term fix as above.)
315 // - Adverse Flight Condition Advisories (weather, equipment, etc.)
316 //   -> [A] hotlines/telcons cover weather, outages, routes (7210.3EE 18-4-1d);
317 //   severe weather management and SWAP are ATCSCC-led (18-15, 18-16).
318 // -> TODO(missing): CTOP route/delay assignment against the operator's TOS (18-12);
```

```
319 // FEA/FCA publication (18-8); Operations Plan / planning webinar (18-21-4a).
320 // These are where TFM constraints first reach the operator's trajectory choice.
321
322 // Core operational coordination - AOC to ATC
323 // - Flight Plan & Route Info (SID/STAR, altitudes, waypoints, etc.)
324 // -> [A] but to a different party: flight plans are filed computer-to-computer
325 // to the ARTCC host via NADIN (7210.3EE 6-5; no earlier than 3 h, phone inside
326 // 45 min). That's aocFlightPlanFiling (to atc), not the ATCSCC link.
327 // - Proposed route modifications
328 // -> [A] Trajectory Options Set (TOS) and Early Intent to TFMS (7210.3EE 18-12-3b,
329 // 18-8-2c); coordination with the ATCSCC Tactical Customer Advocate (18-8-3d).
330 // -
331 // -> TODO(missing): schedule changes/cancellations to FSM (18-10-3); GDP slot
332 // substitution (18-10-12); diversion recovery priorities (18-4-5). Slot
333 // substitution is exactly the airline-vs-system trade the §9 conflict work needs.
334 //
335 // Critique of the whole ATC <-> AOC pair: the old single connect (AOC -> atc) mixed two
336 // different counterparts. The evidence splits it:
337 // strategic/network TFM (EDCT, TMIs, TOS, advisories) -> atcsccl [A, 7210.3EE Ch.18]
338 // flight plan filing / amendment into the host -> atc [A, 7210.3EE 6-5]
339 // The order never says "AOC" or "dispatch", only "system users / flight operators".
340 // Mapping that to airlineOperationsCenter (via its ATC coordinator) is a [B]
341 // abstraction (Seamster p.34, Berry & Pace), recorded in D-009.
342 // RESOLVED (D-009, 2026-09-27): logged as a SPLIT (TFM -> atcsccl, filing stays on atc),
343 // not the pure retarget proposed in the 09-26 journal.
344
345 #ContextVisible connection aocTfm connect flightOps.airlineOperationsCenter to airspaceMgmt.atcsccl {
346 doc /* [A] Traffic flow management / Collaborative Decision Making link. Added 2026-09-27.
347 * ATCSCC -> operator: EDCTs, GDP/GS/AFP/CTOP/reroute advisories, hotline/telcon
348 * content (weather, outages, capacity), CTOP assignments.
349 * Operator -> ATCSCC/TFMS: schedule changes (FSM), TOS, Early Intent, GDP substitutions,
350 * diversion recovery priorities, customer preferences on hotlines.
351 * Party on the AOC side: the ATC coordinator, not the dispatcher (Berry & Pace;
352 * Seamster p.34, p.45).
353 * Authority: the ATCSCC is the "final approving authority" for interfacility TMIs
354 * (18-2-3i); operators must "operate to meet the EDCT" (BB 4-3-4).
355 * Intent chain: flight plan <-> ATC constraints. This is where airline objectives meet
356 * NAS flow objectives (the §9 local-vs-system conflict).
357 * Sources: JO 7210.3EE CHG 3 18-4-1d, 18-8-2c/3d, 18-10-3/6e/12, 18-12-3b, 18-4-5. */
358 }
359 #ContextVisible connection aocFlightPlanFiling connect flightOps.airlineOperationsCenter to airspaceMgmt.atc {
360 doc /* [A] Flight plan filing and amendment. AOC -> ARTCC host (NADIN / computer
361 * interface): filed flight plan with route, SID/STAR, altitudes; amendments. Amended
362 * plan after a TFM reroute is released to HOST (Seamster E-7#14). =?
363 * This was the original AOC -> atc connect; the TFM content moved to aocTfm.
364 * Intent chain: flight plan (the filed intent ATC starts from).
365 * Sources: JO 7210.3EE 6-5, 18-7-4h2; Seamster E-7#14, T2.7#35. */
366 }
367 // TODO(review): `atc` sits beside `tracon`, `artcc`, `atcsccl`, `tmu` in
368 // AirspaceManagementDomain as if it were one more peer, but it's really the umbrella
369 // for tower + TRACON + ARTCC controllers. Fine at context level. When you go to the IBD,
370 // decide whether atc is replaced by those three or holds them as its sub-parts.
371
372 #ContextVisible connection aocAirport connect flightOps.airlineOperationsCenter to airportOps.airport {
373 doc /* [C/B] Airline <-> airport/station coordination.
374 * AOC -> airport: schedules, expected passengers, fuel/catering/baggage needs,
375 * target off-block time (TOBT, A-CDM).
376 * Airport -> AOC: gate assignment, service readiness, boarding status.
377 * Sources: interface-exchange-draft.md (prework hub, C, directions unverified);
378 * Clarke 1998 (B); EUROCONTROL A-CDM (A, but European). */
379 }
380 // TODO(missing): you have no exchange list for AOC <-> Airport, and it's the weakest
381 // link on US evidence. Specifics to fill: (1) gate assignment authority (airline-leased
382 // gates vs airport-assigned); (2) turnaround readiness (fuel, catering, bags) feeding the
383 // off-block time; (3) whether US practice has an A-CDM-like TOBT exchange (FAA TFDM /
384 // JO 7210.637 is the lead; not in the register). Relevance: turnaround readiness sets
385 // the actual departure time, which feeds EDCT compliance and trajectory start.
386
387 #ContextVisible connection aocAircraftDataLink connect flightOps.airlineOperationsCenter to aircraftSys.aircraft {
388 doc /* [B] Added 2026-09-27 from your own note "Flight ops receiving data from the aircraft
389 * systems domain" / "observed by the airline operations center". Aircraft -> AOC
390 * without going through the crew: automated OOOI reports (Off-gate, Off-ground,
391 * On-ground, In-gate), maintenance fault messages, position reports.
392 * AOC -> aircraft: W&B / load uplink, gate uplink, messages to the ACARS printer.
393 * Medium: ACARS / Satcom ("if equipped").
394 * Intent chain: flown trajectory -> AOC (the feedback path the AOC plans against).
395 * Sources: Seamster E-3#10, E-5#06, E-13#12 (automated OOOI), E-2#20, E-10#14;
396 * FAA NAS infrastructure roadmaps (ACARS services). */
397 }
398
```

How will I keep this clean? Really doesn't apply to my objectives, but is involved w/in chain. Does review?

How could I factor in slot changes? DRP. may show valuable exchange w/ decisions important to my work.

Move to placeholder AirspaceMgmt IBD. ☒

☐ speaks to credibility. Look up. equation for delay. Out of scope of project?

How does this fit into what I hope to model, tho?

☒ Research FAA SMART. relation to my project!

```
399 // ATC to Airport Operations
400 // - Runway & Taxiway Status to drive Lights & gate
401 // -> [A] for lights: BB 3-4-2..3-4-15, the tower controller operates the remote
402 // lighting controls. TODO(review): drop "Signage". Taxiway/runway signs are
403 // static, installed and maintained by the airport operator, not driven by ATC.
404 // (Runway Status Lights are automatic from surface surveillance, also not ATC.)
405 // -> TODO(missing) the reverse direction, airport -> ATC: runway/taxiway closures and
406 // field condition / braking reports (airport operator issues field NOTAMs; BB 2-1-9
407 // names the airport manager's office as a NOTAM party); departure/arrival times from
408 // the airport management system ([C] hub draft). Closures change the runway in use,
409 // and with it the SID/STAR/approach, so they do reach the trajectory.
410
411 #ContextVisible connection atcAirport connect airspaceMgmt.atc to airportOps.airport {
412   doc /* [A/C] ATC (tower) <-> airport operator.
413     * ATC -> airport: operation of airfield lighting (BB 3-4-x).
414     * Airport -> ATC: closures, field conditions, NAS component status (BB 2-1-9, C for detail).
415     * Note: the tower is ATC by function but sits physically at the airport
416     * (controlTower is a part of Airport in the definitions). Most tower coordination
417     * is therefore ATC-internal, not this link. The ATC/airport boundary here is movement
418     * area (ATC) vs non-movement area (airport/operator), per the BB glossary and 3-7-2. */
419 }
420
421 // Airport Operations to Crew
422 // - Runway & Taxiway Status via Lights -> [A] BB 3-4-x (lights -> pilot, visual)
423 // -> TODO(missing): ramp/pushback crew <-> captain (push clearance relay, brake/engine-start
424 // cues, marshalling to the gate: Seamster E-3#05-13, E-13#10-13; Wilke 2014). This is
425 // the airport link with real sources behind it; lights alone are the thin part.
426 // -> TODO(review): lights and ramp handling are surface-safety items with weak ties to
427 // trajectory intent. Keep the link for completeness, but don't spend IBD effort here.
428
429 #ContextVisible connection airportToCrew connect flightCrew.flightDeckCrew to airportOps.airport {
430   doc /* [A/B] Airport lighting and markings -> pilot (visual, BB 3-4-x); ramp/pushback
431     * and gate guidance <-> crew (Seamster E-3, E-13; Wilke 2014); fuel slip from
432     * the airport fuel carrier (owner, 2026-09-27). Pushback onto the
433     * movement area needs ATC; who authorizes the push itself varies by airport. */
434 }
435
436 // No exchanges between aircraftsys and airportops, assuming its all visual indicators and not data exchanges.
437 // -> RESOLVED (owner, 2026-09-27): there ARE exchanges, mostly material. They are
438 // kept as a complete set below, but only the summary connection (aircraftAirport)
439 // is rendered. Rationale: none of these exchanges affect optimality. Each one is
440 // driven by another actor (AOC load/fuel orders, passengers, weather, maintenance
441 // status, ATC/ramp clearance), so the decision lever sits on THAT actor's link. The
442 // aircraft <-> airport exchange only carries the decision out.
443 // Visibility rule: the per-exchange connections below have NO #ContextVisible tag,
444 // so MyViews::systemContext skips them. To show one, add the tag.
445 // Endpoints use the Airport / Aircraft sub-parts from the package definitions, so
446 // the set can be reused on an Airport Operations IBD later.
447 //
448 // Complete set: aircraft <-> airport (kind; direction; driven by)
449 // Material, airport -> aircraft
450 //   apFuelUplift      fuel      AOC release fuel order; crew
451 //   apGroundPower     400-HZ electrical  gate assignment; APU policy
452 //   apPreconditionedAir cabin conditioning air  gate assignment; APU policy
453 //   apAirStart        HP air for engine start  MEL (inoperative APU)
454 //   apPotableWater    water service          AOC service plan
455 //   apCatering         galley carts, supplies  AOC passenger count
456 //   apDeicing         de-/anti-icing fluid    weather; crew request
457 //   apLineMaintenance oil, hydraulic fluid,  aircraft status / MEL;
458 //                   O2, N2, tires, parts    maintenance control
459 // Material, both directions
460 //   apPassengers      boarding / deplaning    passengers; AOC schedule
461 //   apBaggage         load / unload          passengers; AOC
462 //   apCargoMail       load / unload          AOC cargo sales
463 //   apWasteService    lav waste, trash       AOC service plan
464 // Physical / mechanical
465 //   apGateDocking     jetway or stairs; chocks  gate assignment (AOC / airport)
466 //   apPushbackTow     tug force; towing        ATC ramp/movement clearance; crew
467 // Information / signals
468 //   apDockingGuidance aircraft position sensed; guidance shown to the crew
469 //                   (VDGS; A-CDM p.36 in-block detection) [C] for US practice
470 //   apRefuelPanel     fueler's preset qty -> aircraft; FQIS qty -> fueler
471 //                   (feeds the fuel slip in PreflightCrossCheck) [C]
472 //   apGroundInterphone ground crew <-> flight deck, plugged into the aircraft
473 //                   (the aircraft is the medium for crew <-> ramp) [B] Seamster E-3
474 //   apGatelinkData    nav database / software loads, maintenance downloads [C]
475 //   apNoiseMonitoring aircraft noise -> airport noise monitors; driven by
476 //                   Governance / community (noiseAbatementProcedures) [C]
477 //   apNavaidSignals   ILS / DME / GBAS -> aircraft nav receivers. The far end
478 //                   is environment.infrastructure, not the airport: D-007
```

☒ Add these content!

☐ Add! Could include stochastic time delay based on flag.

Move to daily note.

so we would need the comms from each actor to airport.

provide D-007 rationale. ☐

```
479 // puts nav aids under Infrastructure, and most ILS are
480 // FAA assets (NAS roadmaps) [B/C]
481 // Evidence (tag in each connection's doc): [B] for the turnaround activities the
482 // registered sources name (fuel, catering, cleaning/waste, boarding/deplaning:
483 // Schultz 2017; bags/cargo: Lu 2025; pushback, interphone: Seamster E-3); [B/C]
484 // nav aids; rest [C] general knowledge (ground power, PCA, air start, water,
485 // de-icing, line maintenance, docking, refuel panel, gatelink, noise). Not driving any
486 // optimization lever, so not worth citing individually unless the report discusses them.
487 // Intentionally left out: lighting/markings (airport -> CREW, on airportToCrew), surface
488 // surveillance (ASDE-X/multilateration: ATC, on atcSurveillance), OOOI (aircraft -> AOC).
489
490 #ContextVisible connection aircraftAirport connect aircraftSys.aircraft to airportOps.airport {
491   doc /* [B] Summary link, rendered in place of the per-exchange set below. Mostly
492     * material turnaround servicing (fuel, power, air, water, catering, bags, cargo,
493     * passengers), gate docking and pushback, plus a few data exchanges (docking
494     * guidance, refuel panel, gatelink). Does not affect optimality: every exchange
495     * executes a decision made on another actor's link (owner, 2026-09-27). */
496 }
497
498 // Per-exchange set (not rendered; see the table above).
499 connection apFuelUplift connect aircraftSys.aircraft.fuelSystem to airportOps.airport.fuelDistributionSystem {
500   doc /* [B] Material, airport -> aircraft: fuel. Driven by the release fuel order and the crew's request. */
501 }
502 connection apGroundPower connect aircraftSys.aircraft to airportOps.airport.gates {
503   doc /* [C] Material, airport -> aircraft: 400 Hz ground power at the gate (or a GPU cart). */
504 }
505 connection apPreconditionedAir connect aircraftSys.aircraft to airportOps.airport.gates {
506   doc /* [C] Material, airport -> aircraft: preconditioned cabin air at the gate. */
507 }
508 connection apAirStart connect aircraftSys.aircraft to airportOps.airport.groundHandling {
509   doc /* [C] Material, airport -> aircraft: high-pressure air for engine start. Needed when the APU is inoperative
(MEL). */
510 }
511 connection apPotableWater connect aircraftSys.aircraft to airportOps.airport.groundHandling {
512   doc /* [C] Material, airport -> aircraft: potable water service. */
513 }
514 connection apCatering connect aircraftSys.aircraft to airportOps.airport.groundHandling {
515   doc /* [B] Material, airport -> aircraft: galley carts and supplies, sized by the AOC's passenger count. Caterer
modeled under groundHandling. */
516 }
517 connection apDeicing connect aircraftSys.aircraft to airportOps.airport.groundHandling {
518   doc /* [C] Material, airport -> aircraft: de-/anti-icing fluid. Driven by weather and the crew's request. */
519 }
520 connection apLineMaintenance connect aircraftSys.aircraft to airportOps.airport.maintenanceHangar {
521   doc /* [C] Material, both directions: servicing fluids and gases, tires and parts in; removed parts out. Driven by
aircraft status / MEL via maintenance control. Line maintenance happens at the gate; maintenanceHangar is the only maintenance
part in the definitions. */
522 }
523 connection apPassengers connect aircraftSys.aircraft to airportOps.airport.passengerTerminal {
524   doc /* [B] Material, both directions: boarding and deplaning via the gate / jetway. */
525 }
526 connection apBaggage connect aircraftSys.aircraft to airportOps.airport.baggageHandlingSystem {
527   doc /* [B] Material, both directions: checked baggage loaded and unloaded. */
528 }
529 connection apCargoMail connect aircraftSys.aircraft to airportOps.airport.cargoTerminal {
530   doc /* [B] Material, both directions: cargo and mail loaded and unloaded. */
531 }
532 connection apWasteService connect aircraftSys.aircraft to airportOps.airport.groundHandling {
533   doc /* [B] Material, aircraft -> airport: lavatory waste and cabin trash; cabin cleaning. */
534 }
535 connection apGateDocking connect aircraftSys.aircraft to airportOps.airport.gates {
536   doc /* [C] Physical: jetway or stairs attachment, chocks, parking position. */
537 }
538 connection apPushbackTow connect aircraftSys.aircraft to airportOps.airport.groundHandling {
539   doc /* [B] Physical: tug pushback and towing. Driven by ATC / ramp clearance and the crew. */
540 }
541 connection apDockingGuidance connect aircraftSys.aircraft to airportOps.airport.gates {
542   doc /* [C] Information: the visual docking guidance system senses the aircraft's position and shows stop guidance
to the crew; can also detect in-block / off-block (A-CDM p.36). */
543 }
544 connection apRefuelPanel connect aircraftSys.aircraft.fuelSystem to airportOps.airport.fuelDistributionSystem {
545   doc /* [C] Information, both directions: fueler presets quantity at the refuel panel; aircraft fuel quantity read
back to the fueler. Source of the fuel slip. */
546 }
547 connection apGroundInterphone connect aircraftSys.aircraft to airportOps.airport.groundHandling {
548   doc /* [B] Information: ground crew headset plugged into the aircraft interphone. The aircraft is the medium for
crew <-> pushback crew (Seamster E-3#05-13). */
549 }
550 connection apGatelinkData connect aircraftSys.aircraft to airportOps.airport.gates {
551   doc /* [C] Information, both directions: navigation database and software loads in; maintenance and flight data
downloads out.*/
```

```
552 }
553 connection apNoiseMonitoring connect aircraftSys.aircraft to airportOps.airport {
554     doc /* [C] Information, aircraft -> airport: noise measured by airport monitors against
Airport.noiseAbatementProcedures. Driven by Governance / community. */
555 }
556 connection apNavaidSignals connect aircraftSys.aircraft.navigationSystem to environment.infrastructure {
557     doc /* [B/C] Information, infrastructure -> aircraft: ILS localizer/glideslope, DME, GBAS signals. Physically at
the airport, but D-007 places navaids in Infrastructure, and most ILS are FAA assets (NAS roadmaps). */
558 }
559
560 // Aircraft Systems to Crew
561 // - Aircraft & Engine Status -> [C] flight-deck displays/alerting
562 // - Fuel Status -> [C]
563 // - Navigation & Avionics Status -> [C]
564 // - Equipment Malfunctions & Emergencies -> [C] crew alerting
565 // - Position & Trajectory Info (via FMS) -> [C] ND/PFD
566 // - Recommended Aircraft Control (via FMS: speed, altitude, heading, etc.)
567 // -> [C] flight director / autopilot guidance. This is the "FMS intent -> guidance
568 // commands" step of the intent chain, so it's central, not peripheral.
569 // -> TODO(missing) the crew -> aircraft direction, which is the actual execution
570 // step: control inputs, MCP/FCU selections, FMS entries (route, SID/STAR, weights,
571 // fuel, cost index: ConOps Phase 3), configuration (flaps, gear).
572 // -> TODO(review) evidence: all [C]. Needs an FCOM / FAA AC citation, or the avionics
573 // sources already registered (GreAT D5.1, Mhenni 2016) before the report cites it.
574 // Intuitively solid, so the connection stays.
575
576 #ContextVisible connection crewAircraft connect flightCrew.flightDeckCrew to aircraftSys.aircraft {
577     doc /* [C] Crew <-> aircraft execution.
578         * Aircraft -> crew: status/alerting, fuel quantity (FQIS gauges), nav/avionics status,
579         * position and trajectory, FMS performance predictions (fuel at destination, ETAs,
580         * gross weight, optimum altitude) the crew checks against the release,
581         * FMS guidance (flight director / autopilot targets).
582         * Crew -> aircraft: control inputs, MCP/FCU selections, FMS programming, configuration.
583         * PF/PM duties swap between the two pilots, so the flight deck is one end.
584         * Intent chain: FMS intent -> guidance commands -> aircraft motion. */
585 }
586
587 #ContextVisible connection atcSurveillance connect airspaceMgmt.atc to aircraftSys.aircraft {
588     doc /* [C] Surveillance, not command. Aircraft -> ATC: transponder (Mode C/S) replies and
589         * ADS-B broadcasts; radar returns feed tower/TRACON/ARTCC displays (data block,
590         * handoff status). ATC -> aircraft: radar interrogation only. Instructions go through
591         * the crew (atcToCrew). The gap between the aircraft's true state and ATC's tracked
592         * picture is the cyber-physical gap (Mordecai 2018).
593         * Intent chain: aircraft motion -> ATC (the feedback that closes the loop). */
594 }
595 // TODO(review) evidence: [C] only (your 09-20 lecture capture). BB's beacon code
596 // and ADS-B paragraphs would make this [A]; not yet extracted.
597
598 // Inter-ATC Connections
599 // - ATSCC to TMU:
600 // - Flow Evaluation Areas (FEAs) and Flow Constrained Areas (FCAs). This lets the TMU know exactly which flights
in their sectors are impacted by macro-level constraints
601 // -> [A] 7210.3EE 18-8-4a3, 18-7-13. TODO(review) term: "ATSCC" -> "ATCSCC"
602 // (and the model part is atcsc).
603 // - Traffic Management Initiatives (Ground Delay Programs (GDPs) and Ground Stops (GS): the TMU coordinates
Expect Departure Clearance Times (EDCTs))
604 // -> [A] with a correction: the ATCSCC ISSUES EDCTs (18-10-6e). The ARTCC TMU
605 // relays them, assigns from the DAS table in some modes, and requests revised
606 // EDCTs from the ATCSCC (18-10-7). TODO(review): reword "coordinates".
607 // - Flight Schedule Monitor (FSM): Used to transmit slot assignments, capacity limits, and real-time modeling
data to TMU terminals
608 // -> TODO(review): FSM is a TOOL (a medium), not an exchange. Operators use it too
609 // (18-10-3). The exchange items are the slot assignments / capacity (arrival
610 // rates). List those here and treat FSM as the channel.
611 // - TMU to ATSCC:
612 // - Disruption Events/Changes (localized convective weather, low visibility, or wind changes that call for pre-
coordinated Severe Weather Avoidance Plans (SWAP), runway closures, airport equipment failures, computer/software glitches,
etc.)
613 // -> [A] 18-2-4a6, 18-5-4c. Nuance: SWAP is ATCSCC-led (18-16). The TMU reports
614 // the conditions; the SWAP decision is the ATCSCC's.
615 // - MITs, including submitting or updating Miles-in-Trail (MIT) or Minutes-in-Trail restrictions between
adjacent ARTCC sectors
616 // -> [A] with nuance (18-7-12/13): up to 10 MIT is coordinated facility to facility;
617 // 25+ MIT needs a shared FEA; 40+ MIT needs a conference. So the ATCSCC sees the
618 // large ones, not all of them.
619 // - TBFM
620 // -> TODO(review): TBFM is a tool, like FSM. What goes to the ATCSCC is time-based
621 // management STATUS (18-5-4c1, 18-6-9b, 18-6-11). TBFM metering is the TBO link
622 // (18-6-1) closest to trajectory intent, so it's worth stating precisely.
623 // -> TODO(missing): delay reports (15-min verbal, NTML updates; 18-5-12); reroute
624
```

ADD.
 Try to find or explain intuition from work. SME ref to bib?

Change on ref?
 PIC/PF Captain/Bo

check for source.

change!

change!

Review & Update

Update!
 What is NTML??

```
625 // requests to the ATCSCC via NTML (18-5-11). Medium for almost all of this: NTML
626 // "in preference to other methods" (18-5-3) plus hotlines.
627 // -> TODO(missing): the TMU <-> tactical ATC link (tower/TRACON/ARTCC controllers). That
628 // is how a TMI becomes a clearance (BB 4-3-4: tower checks EDCTs with the TMU, and
629 // TMUs pass MIT/metering to sectors). Without it the diagram doesn't show how TFM
630 // constraints reach the crew. Draft below.
631 // -> TODO(review) scope: ATCSCC <-> TMU is internal to Airspace Management. On a strict
632 // context diagram, links internal to one domain are out of scope; they belong on the
633 // Airspace Management IBD (§10). I kept it visible because it's the enterprise ->
634 // tactical step of the intent chain. Decide whether the context view keeps it.
635
636 #ContextVisible connection atcscctmu connect airspaceMgmt.atcscctmu to airspaceMgmt.tmu {
637   doc /* [A] National <-> local traffic flow management. Added 2026-09-27 from your notes.
638   * ATCSCC -> TMU: FEA/FCA, TMIs (GDP, GS, AFP, CTOP, reroutes), EDCTs, slot/rate data (via FSM).
639   * TMU -> ATCSCC: component outages and disruptions, weather conditions, MIT >= 25
640   * (FEA) / >= 40 (conference), TBM status, delay reports, reroute requests.
641   * Medium: NTML (preferred), TFMS, FSM, hotlines/telcons.
642   * Authority: ATCSCC is the final approving authority for interfacility TMIs (18-2-3i);
643   * TMUs keep local latitude (sector MIT, fix balancing; 18-2-3 NOTE).
644   * Hierarchy the single tmu part stands for: tower -> TRACON TMU -> ARTCC TMU -> ATCSCC.
645   * Intent chain: enterprise/NAS flow objectives -> ATC constraints.
646   * Sources: JO 7210.3EE CHG 3 Ch. 18. */
647 }
648
649
650 // =====
651 // MISSING: draft code for TODO(missing) items. Uncomment to include.
652 // =====
653
654 // --- TMU <-> tactical ATC. [A] BB 4-3-4 (tower <-> TMU on EDCT, ATCSCC via TMU);
655 // 7210.3EE 18-2-4 (TMUs direct flows in conjunction with the ATCSCC). This is how a
656 // TMI turns into the clearance the crew receives.
657 // #ContextVisible connection tmuatc connect airspaceMgmt.tmu to airspaceMgmt.atc {
658 //   doc /* TMU -> controllers: MIT/MINIT, metering times (TBFM), EDCT verification,
659 //   * departure release. Controllers -> TMU: sector load, pilot-reported EDCT checks,
660 //   * departure readiness outside the EDCT window. */
661 // }
662
663 // --- Cabin crew <-> PIC. [C] cabin secure / ready for departure; passenger medical or
664 // security events that trigger a diversion (Seamster T2.10 covers diversion on the
665 // crew <-> AOC side). Only needed if cabinCrew stays on the diagram.
666 // #ContextVisible connection cabinToFlightDeck connect flightCrew.cabinCrew to flightCrew.flightDeckCrew {
667 //   doc /* Cabin -> PIC: cabin secure, passenger/medical/security events.
668 //   * PIC -> cabin: phase-of-flight and turbulence advisories. */
669 // }
670
671 // --- Environment / boundary actors. environment is declared but not visible or connected.
672 // The evidence note proposes three kinds of flow: rules (static), disturbances
673 // (dynamic scenario inputs), and feedback. Link an actor only where it changes a
674 // decision. To use: replace `part environment: EnvironmentDomain;` at the top with
675 // the block below, then uncomment the connections you want.
676 // #ContextVisible part environment: EnvironmentDomain {
677 //   #ContextVisible part :>> governance;
678 //   #ContextVisible part :>> military;
679 //   #ContextVisible part :>> infrastructure {
680 //     #ContextVisible part :>> weatherService;
681 //   }
682 // }
683 // // Rules [A]: 14 CFR 91/121, JO 7110.65BB / 7210.3EE bind the operator and ATC.
684 // #ContextVisible connection rulesToAoc connect environment.governance to flightOps.airlineOperationsCenter;
685 // #ContextVisible connection rulesToAtc connect environment.governance to airspaceMgmt.atc;
686 // // Disturbance [A]: weather drives TMIs and SWAP (7210.3EE 18-15/16), ATIS (BB 2-9)
687 // // and dispatch planning.
688 // #ContextVisible connection weatherToTfm connect environment.infrastructure.weatherService to airspaceMgmt.atcscctmu;
689 // #ContextVisible connection weatherToAoc connect environment.infrastructure.weatherService to
flightOps.airlineOperationsCenter;
690 // // Disturbance [C]: special-use airspace status as a scenario condition (D-007).
691 // #ContextVisible connection suaToTfm connect environment.military to airspaceMgmt.atcscctmu;
692 // // Signal: nav aids -> aircraft now exists, unrendered, as apNavAidSignals (aircraft <->
693 // // airport set). To show it, add #ContextVisible there rather than adding a second link.
694
695 // --- Preflight plan-vs-FMS-vs-fuel cross-check (owner intent 2026-09-27; [B] for the
696 // release and fuel-on-board report, [C] for the cross-check itself). The dispatch
697 // release is the item that carries the plan; the crew compares it with what the FMS
698 // predicts and with the fuel actually loaded, and only the outcome goes back to the
699 // AOC. Written as items + an action def because the comparison happens inside the
700 // crew, not on a link. To use: move the item defs and action def into
701 // nas_sysml_package_definitions.sysml (methods §10.1: defs live there, this file
702 // holds usages and wiring only); then give the action to flightDeckCrew here
703 // (e.g. `perform action preflightCrossCheck: PreflightCrossCheck;`).
```

How to add? TMI only?

hmmmm... #TODO

"order"

REVIEW

```
704 // item def DispatchRelease {
705 //   doc /* The dispatch paperwork: route, planned burn and times, alternates,
706 //     * minimum/planned fuel, planned ZFW and takeoff weight, cost index, MEL items,
707 //     * weather/NOTAMs. Signed by dispatcher and captain. Delivered on paper, EFB or ACARS. */
708 //   attribute plannedFuel: Real;           // lb
709 //   attribute plannedBurn: Real;           // lb
710 //   attribute plannedZFW: Real;           // lb
711 //   attribute plannedTakeoffWeight: Real; // lb
712 //   attribute costIndex: Real;
713 //   attribute v1: Real;                    // kt; takeoff performance modeled as AOC-provided
714 //   attribute vR: Real;                    // kt  (abstraction; some types compute these in
715 //   attribute v2: Real;                    // kt   the FMS, e.g. 737)
716 //   attribute takeoffThrustSetting: String; // e.g. derate / assumed temperature
717 // }
718 // item def FuelSlip {
719 //   doc /* Airport fuel carrier's uplift record: gallons delivered and density, converted to pounds. */
720 //   attribute gallonsUplifted: Real;
721 //   attribute density: Real;              // lb/gal
722 // }
723 // item def FmsPerformancePrediction {
724 //   doc /* FMS-computed values from the crew-entered ZFW, fuel, cost index, winds. */
725 //   attribute predictedFuelAtDestination: Real; // lb
726 //   attribute predictedGrossWeight: Real;       // lb
727 //   attribute optimumAltitude: Real;            // ft
728 // }
729 // item def PreflightCrossCheckResult {
730 //   doc /* What goes crew -> AOC: accepted, or a discrepancy needing an amended release. */
731 //   attribute fuelOnBoard: Real;                // lb, from gauges, reconciled against the slip
732 //   attribute accepted: Boolean;
733 //   attribute discrepancy: String;              // e.g. "fuel", "weight", "performance limit"
734 // }
735 // action def PreflightCrossCheck {
736 //   in release: DispatchRelease;                // via aocToCrew
737 //   in fuelSlip: FuelSlip;                      // from the airport fuel carrier, via airportToCrew
738 //   in fmsPrediction: FmsPerformancePrediction; // via crewAircraft
739 //   in gaugeFuel: Real;                         // FQIS, via crewAircraft
740 //   out result: PreflightCrossCheckResult;      // to AOC via aocToCrew
741 // }
742
743 // --- Directed, typed exchanges (the next step past undirected `connection`s).
744 // Everything above is an undirected link with its content in a doc comment. For §10
745 // (item-flow model), the content becomes item defs moving through ports. Worked
746 // example on the A-graded AOC <-> ATCSCC link; the same pattern repeats per link.
747 // The evidence note suggests port types by MEDIUM (radio, ACARS, phone/telcon,
748 // NTML) rather than one per pair. The two ports below are conjugates (~) of one def.
749 // To use: move the item defs and port def into nas_sysml_package_definitions.sysml
750 // (methods §10.1), add the `port` lines inside the existing airlineOperationsCenter /
751 // atcsccl redefinitions at the top, then uncomment the flows here.
752 // item def EDCT {
753 //   doc /* Expect Departure Clearance Time (JO 7210.3EE 18-10-6e). */
754 //   attribute flightNumber: String;
755 // }
756 // item def TrajectoryOptionsSet {
757 //   doc /* Operator-ranked route options for CTOP (7210.3EE 18-12-3b). */
758 // }
759 // item def TMIAdvisory {
760 //   doc /* GDP / GS / AFP / CTOP / reroute advisory (18-10-6c). */
761 // }
762 // port def TFMCustomerPort {
763 //   in item edct: EDCT;
764 //   in item advisory: TMIAdvisory;
765 //   out item tos: TrajectoryOptionsSet;
766 // }
767 // inside flightOps { part :>> airlineOperationsCenter { port tfm: TFMCustomerPort; } }
768 // inside airspaceMgmt { part :>> atcsccl { port customers: ~TFMCustomerPort; } }
769 // #ContextVisible flow of EDCT from airspaceMgmt.atcsccl.customers.edct to flightOps.airlineOperationsCenter.tfm.edct;
770 // #ContextVisible flow of TMIAdvisory from airspaceMgmt.atcsccl.customers.advisory to flightOps.airlineOperationsCenter.tfm.advisory;
771 // #ContextVisible flow of TrajectoryOptionsSet from flightOps.airlineOperationsCenter.tfm.tos to
772 // airspaceMgmt.atcsccl.customers.tos;
773 }
774 }
775
```